

The diagonal sampled second-order Riesz transform is unbounded

A counterexample to Conjecture 1.4 of arXiv:2504.18739

Candidate counterexample packet

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Status: counterexample_likely_valid.

1 Exact target

Bañuelos and Kim define, for $d \geq 2$ and $1 < p < \infty$, the sampled second-order transforms on $\ell^p(\mathbb{Z}^d)$ by

$$R_{\text{dis}}^{(jk)} f(n) = \sum_{m \in \mathbb{Z}^d} K_{\text{dis}}^{(jk)}(m) f(n - m), \quad K_{\text{dis}}^{(jk)}(m) = c_d \frac{m_j m_k}{|m|^{d+2}} \mathbf{1}_{\mathbb{Z}^d \setminus \{0\}}(m), \quad (1)$$

where $c_d = \pi^{-d/2} \Gamma((d+2)/2) > 0$ [1, (1.16)–(1.17)]. The definition is reproduced verbatim below.

in the area commonly known as Discrete Analogues in Harmonic Analysis (DAHA), which concerns the study of discrete counterparts on $\mathbb{Z}^d$ of operators in harmonic analysis on $\mathbb{R}^d$, we refer the reader to [33, 38, 41, 42].

In this paper, motivated by the recent results in [5, 6, 11] concerning sharp estimates for DAHA of the Hilbert and first-order Riesz transforms, we investigate similar problems for the DAHA of second-order Riesz transforms defined as in (1.14) by

$$(1.16) \quad R_{\text{dis}}^{(jk)}(f)(n) = \sum_{m \in \mathbb{Z}^d} K_{\text{dis}}^{(jk)}(m) f(n - m),$$

where

$$(1.17) \quad K_{\text{dis}}^{(jk)}(m) = c_d \frac{m_j m_k}{|m|^{d+2}} \mathbf{1}_{\mathbb{Z}^d \setminus \{0\}}(m), \quad c_d = \pi^{-\frac{d}{2}} \Gamma\left(\frac{d+2}{2}\right).$$

Notice that since $|K_{\text{dis}}^{(jk)}(m)| \leq \frac{c_d}{|m|^d}$, it follows from Hölder's inequality that the sum in (1.16) is absolutely convergent for $f \in \ell^p$, $1 < p < \infty$. Similarly, absolute convergence holds for the general discrete singular integrals with Calderón-Zygmund kernels as in (1.13) and even more general singular integrals of non-convolution type [28, Chapter 8].

Conjecture 1.4 then asserts, among other statements, that for every diagonal index j ,

$$\|\mathcal{R}^{(jj)}\|_{\ell^p \rightarrow \ell^p} = \|R_{\text{dis}}^{(jj)}\|_{\ell^p \rightarrow \ell^p} = \|R^{(jj)}\|_{L^p \rightarrow L^p}. \quad (2)$$

Remark 1.3. We observe that the second equalities in (1.18), (1.19) and (1.20) are in fact the equalities in (1.7), (1.8) and (1.9). It will be shown in Section 3 (Corollary 3.1), that “ $\leq$ ” in the first equalities of (1.18), (1.19), as well as the inequality in (1.20), follow from the martingale inequalities of Burkholder and Choi. The proof of the opposite inequalities in (1.18) and (1.19) is done in Propositions 4.1 and 4.4 using a dilation and approximation argument. Such an argument does not apply to the kernels for $\mathcal{R}^{(jj)}$; see the comments in the introduction of Section 4 and Remark 4.2.

Conjecture 1.4. For all $j, k = 1, 2, \dots, d$, $j \neq k$,

$$(1.22) \quad \|\mathcal{R}^{(jk)}\|_{\ell^p \rightarrow \ell^p} = \|R_{\text{dis}}^{(jk)}\|_{\ell^p \rightarrow \ell^p} = \|R^{(jk)}\|_{L^p \rightarrow L^p},$$

and

$$(1.23) \quad \|\mathcal{R}^{(jj)} - \mathcal{R}^{(kk)}\|_{\ell^p \rightarrow \ell^p} = \|R_{\text{dis}}^{(jj)} - R_{\text{dis}}^{(kk)}\|_{\ell^p \rightarrow \ell^p} = \|R^{(jj)} - R^{(kk)}\|_{L^p \rightarrow L^p}.$$

For all $j = 1, 2, \dots, d$,

$$(1.24) \quad \|\mathcal{R}^{(jj)}\|_{\ell^p \rightarrow \ell^p} = \|R_{\text{dis}}^{(jj)}\|_{\ell^p \rightarrow \ell^p} = \|R^{(jj)}\|_{L^p \rightarrow L^p}.$$

Notice that (1.7), (1.8), and the Calderón-Zygmund inequality (1.15) give

$$(1.25) \quad \begin{aligned} \|2R_{\text{dis}}^{(jk)}\|_{\ell^p \rightarrow \ell^p} &\leq \|2R^{(jk)}\|_{L^p \rightarrow L^p} + C_d \\ &= (p^* - 1) + C_d, \quad j \neq k, \end{aligned}$$

and

$$(1.26) \quad \begin{aligned} \|R_{\text{dis}}^{(jj)} - R_{\text{dis}}^{(kk)}\|_{\ell^p \rightarrow \ell^p} &\leq \|R^{(jj)} - R^{(kk)}\|_{L^p \rightarrow L^p} + C_d \\ &= (p^* - 1) + C_d, \quad i \neq k. \end{aligned}$$

The diagonal sampled kernel in (1) is nonnegative and has no cancellation. The next theorem shows that its norm is not merely larger than the conjectured value: it is infinite.

2 Counterexample theorem

For $N \in \mathbb{N}$, write

$$Q_N = [-N, N]^d \cap \mathbb{Z}^d.$$

Theorem 1. Fix $d \geq 2$ and $j \in \{1, \dots, d\}$. The convolution operator on finitely supported sequences with kernel

$$k_j(m) = c_d \frac{m_j^2}{|m|^{d+2}} \mathbf{1}_{\mathbb{Z}^d \setminus \{0\}}(m)$$

has no bounded extension to $\ell^p(\mathbb{Z}^d)$ for any $1 < p \leq \infty$. Consequently the diagonal assertion (2), and hence Conjecture 1.4 as stated, is false. Already for $p = 2$,

$$\|R_{\text{dis}}^{(jj)}\|_{\ell^2 \rightarrow \ell^2} = \infty \quad \text{while} \quad \|R^{(jj)}\|_{L^2 \rightarrow L^2} = \gamma(2) = 1.$$

Proof. Put

$$a_d = c_d(d+3)^{-(d+2)/2} > 0.$$

For an integer $L \geq 1$, set $N = 2^{L+1}$ and $f_N = \mathbf{1}_{Q_{2N}}$. For $q = 0, \dots, L-1$, let $r_q = 2^q$ and define the disjoint lattice sets

$$S_q = \left\{ m \in \mathbb{Z}^d : r_q \leq m_j < 2r_q, \quad 0 \leq m_i < r_q \ (i \neq j) \right\}.$$

They satisfy $|S_q| = r_q^d$ and $S_q \subseteq Q_{N/2}$. If $m \in S_q$, then

$$m_j^2 \geq r_q^2, \quad |m|^2 \leq (4 + d - 1)r_q^2 = (d + 3)r_q^2,$$

and therefore

$$k_j(m) \geq a_d r_q^{-d}. \quad (3)$$

Fix $n \in Q_N$. For $m \in S_q \subseteq Q_{N/2}$ we have $n - m \in Q_{3N/2} \subseteq Q_{2N}$, so $f_N(n - m) = 1$. Positivity of k_j , disjointness of the S_q , and (3) give

$$(R_{\text{dis}}^{(jj)} f_N)(n) \geq \sum_{q=0}^{L-1} \sum_{m \in S_q} k_j(m) \geq \sum_{q=0}^{L-1} r_q^d a_d r_q^{-d} = a_d L.$$

Thus, for $1 < p < \infty$,

$$\begin{aligned} \frac{\|R_{\text{dis}}^{(jj)} f_N\|_{\ell^p}}{\|f_N\|_{\ell^p}} &\geq a_d L \left(\frac{|Q_N|}{|Q_{2N}|} \right)^{1/p} \\ &= a_d L \left(\frac{2N+1}{4N+1} \right)^{d/p} \geq a_d 2^{-d/p} L \longrightarrow \infty. \end{aligned}$$

For $p = \infty$, the same pointwise estimate gives a ratio at least $a_d L$. This proves unboundedness. The last claim follows from the finite classical norm recorded in [1, (1.9), (1.24)], with $\gamma(2) = 1$. $\square$

3 Why the literal definition leaves no summation escape

The test above uses only finitely supported f_N . At every n , the sum in (1) then contains only finitely many nonzero terms. Hence no choice of principal value, enumeration, or convergence convention can alter the estimate. Adding a compensating local term or replacing $f(n - m)$ by $f(n - m) - f(n)$ would define a different operator from (1.16)–(1.17).

The issue is the familiar cancellation obstruction for degree- $-d$ kernels. For $j \neq k$, the angular factor $m_j m_k$ changes sign. For the diagonal sample it becomes $m_j^2 \geq 0$ and has positive mean on every fixed cone, so each dyadic scale contributes a positive constant. Classical diagonal second-order Riesz transforms are distributional singular integrals with a local compensation; simply deleting the origin and sampling the positive off-origin function loses that compensation.

Corollary 2. *Under the literal definitions of [1], the diagonal instance of the comparison (1.21), the diagonal bound (3.15), and the all-index identity*

$$\mathcal{R}^{(jk)} + R_{\text{dis}}^{(jk)} = \mathcal{J}^{(jk)} * \quad \text{with } \mathcal{J}^{(jk)} \in \ell^1$$

cannot all be valid for $j = k$.

Proof. The paper's martingale construction bounds $\mathcal{R}^{(jj)}$ on ℓ^p , and convolution by an ℓ^1 sequence is bounded on ℓ^p . The displayed identity would therefore make $R_{\text{dis}}^{(jj)}$ bounded, contradicting Theorem 1. $\square$

4 Audit of the diagonal proof

The counterexample is independent of the source proof, but that proof shows where the diagonal clause entered. On PDF page 20, the integral of a product of two derivative factors is separated into the factor

$$F_2(m_j, t)F_3(m_k, t)$$

times the remaining coordinate factors. That separation is available when $j \neq k$. When $j = k$, both derivatives live in the same coordinate, so their product must remain inside one one-dimensional integral; the displayed product counts that coordinate twice.

There is also a visible sign failure on PDF page 21: a strictly positive integral of

$$\frac{P_t^{(1)}(x)P_t^{(1)}(x-m)}{\theta(x, t)}$$

is displayed as its own negative after a change of variables, and is then declared zero. Since each factor is positive, that conclusion cannot hold. The relevant source passage is reproduced below.

We will first simplify $F_2(m_j, t)$ and $F_3(m_k, t)$. By the change of variable $y = m_k - x_k$ and the symmetry of $P_t^{(1)}(x_k)$ in x_k , we have

$$\begin{aligned} \int_{\mathbb{R}} \frac{P_t^{(1)}(x_k)P_t^{(1)}(x_k - m_k)}{\theta(x_k, t)} dx_k &= - \int_{\mathbb{R}} \frac{P_t^{(1)}(m_k - y)P_t^{(1)}(-y)}{\theta(m_k - y, t)} dy \\ &= - \int_{\mathbb{R}} \frac{P_t^{(1)}(y - m_k)P_t^{(1)}(y)}{\theta(m_k - y, t)} dy. \end{aligned}$$

Since $\theta(m_k - y, t) = \theta(y, t)$ by (2.6), the integral is zero and so

$$\int_{\mathbb{R}} \frac{(x_k - m_k)P_t^{(1)}(x_k)P_t^{(1)}(x_k - m_k)}{t\theta(x_k, t)} dx_k = \int_{\mathbb{R}} \frac{x_k P_t^{(1)}(x_k)P_t^{(1)}(x_k - m_k)}{t\theta(x_k, t)} dx_k.$$

For $l = j, k$, the change of variable $y = m_l/2 - x$ yields

$$\int_{\mathbb{R}} P_t^{(1)}(x)P_t^{(1)}(x - m_l) \frac{\partial_z \theta(x, t)}{\theta(x, t)^2} dx = - \int_{\mathbb{R}} P_t^{(1)}\left(\frac{m_l}{2} - y\right) P_t^{(1)}\left(\frac{m_l}{2} + y\right) \frac{\partial_z \theta(\frac{m_l}{2} - y, t)}{\theta(\frac{m_l}{2} - y, t)^2} dy$$

Since $P_t^{(1)}(\frac{m_l}{2} - y)P_t^{(1)}(\frac{m_l}{2} + y)$ and $\theta(\frac{m_l}{2} - y, t)$ are even functions in y , the integrand is odd. So, we have

$$\int_{\mathbb{R}} P_t^{(1)}(x)P_t^{(1)}(x - m_l) \frac{\partial_z \theta(x, t)}{\theta(x, t)^2} dx = 0.$$

Even if the latter line is treated as a typographical sign error, the $j = k$ factorization failure and the direct box counterexample remain.

5 Scope and upgrade audit

The result is a full counterexample to Conjecture 1.4 because the conjecture is a conjunction containing the universal diagonal clause (1.24). The proof strengthens the minimal $d = 2$, $p = 2$ refutation to every

$d \geq 2$, every diagonal index, and every $1 < p \leq \infty$. At $p = 1$, the image of a point mass already fails to lie in ℓ^1 , since the positive kernel has divergent total mass.

This packet does *not* settle the off-diagonal equality (1.22), the trace-free diagonal-difference equality (1.23), or the Beurling–Ahlfors conjecture. Those kernels retain cancellation and are not touched by the positive-cone argument. A natural repair is to restrict the sampled-kernel program to off-diagonal and trace-free combinations, or to define a genuinely compensated diagonal discretization.

Eight focused checks were made: literal source scope; the minimal $d = 2, p = 2$ test; extension across p ; extension across d ; endpoint behavior; the bounded-plus- ℓ^1 consistency contradiction; localization of the source proof failure; and bounded novelty/surviving-scope searches. Searches of the run’s four lightweight indexes and by arXiv id, exact title, conjecture number, and kernel formula found no correction, erratum, or prior counterexample. This is therefore a candidate result awaiting expert review, not a definitive priority claim.

Human-review recommendation

Check first that equations (1.16)–(1.17) are being read literally and that no unwritten compensation convention is intended for $j = k$. Under the displayed definition, the dyadic-box proof is elementary and decisive. A reviewer should then compare the $j = k$ case against the published authors’ intended operator and decide whether the correct response is a counterexample to (1.24) or an erratum to the diagonal definition.

References

- [1] R. Bañuelos and D. Kim, *Discrete analogues of second-order Riesz transforms*, arXiv:2504.18739v2 (2026); Conjecture 1.4.